

Workshop Activities

Activity 1 – Evaluating the Design of a Data Dashboard

An alternative version of the Espléndido Jugo y Batido data dashboard is shown below. How would you modify this alternative to improve the dashboard?



Activity 2 – Adding a Chart to a Data Dashboard

In addition to the information provided on the data dashboard shown below, Espléndido Jugo y Batido management has decided it would like to include a series of charts that show the relationship between average service satisfaction rating and average product satisfaction rating across the distribution centres for each of the past three years. EJB Management would also like to include a table that provides average service satisfaction rating and average product satisfaction rating by distribution centre across years.



- What type of chart(s) would you recommend to show the relationship between the average of the service satisfaction rating and the average of product satisfaction rating across the distribution centres for each of the past three years? Explain why you would suggest such chart(s).
- How would you revise the data dashboard to incorporate the chart or charts you suggested in part a?

Activity 3 – Key Performance Indicators

Consider the below list of bullet points providing common key performance indicators (KPIs) for manufacturing dashboards:

- the quantity produced
- the quality of units produced
- the units' production speed
- the frequency of problems (machine downtime, returns, falling short of production targets, etc.)

Now suggest other KPIs that would help a manufacturing centre to function effectively.

Activity 4 – Building a Data Dashboard using Excel

EJB wants to develop a data dashboard that its mid-level operations management team can use to track its sales over the most recent three years. The objectives for the data dashboard have been established.

EJB wants:

- to be able to track dollar sales for each of its distribution centres by year and it wants to be able to look at these data for new and existing customers,
- to be able to track dollar sales by year for each category (juices and smoothies) and it would like to be able to look at these data by flavour and by month
- to be able to monitor time to deliver by distribution centre for each year and it wants to be able to generate sales in dollars across category and flavour by distribution centre at the year, month and date ordered levels, and
- the data dashboard to automatically update when new data are added to the source data.

The KPIs for the dashboard are total dollar sales and average delivery times. In addition, EJB's requirement to be able to specify the month, year or date ordered, the category, flavour, distribution centre and whether new or existing customers are displayed in various charts suggests that a dynamic and interactive data dashboard is required.

The functionality that EJB wants in its data dashboard can be provided through the following charts and tables:

- a stacked column chart of total sales across distribution centres and year ordered by new or existing customers
- a line chart of total sales across year and month ordered by new or existing customers
- a clustered column chart of total sales across year ordered and flavour by category
- a clustered bar chart of average time to deliver across distribution centres by year ordered
- a table of total sales across category and flavour by distribution centre and year, month, and day ordered

The *EJBChart1* data file provides the *EJBData* Excel Table in the Data worksheet and the first chart with slicers in the Chart1 worksheet. PivotTables, PivotCharts and slicers can be used to build the remaining three charts.

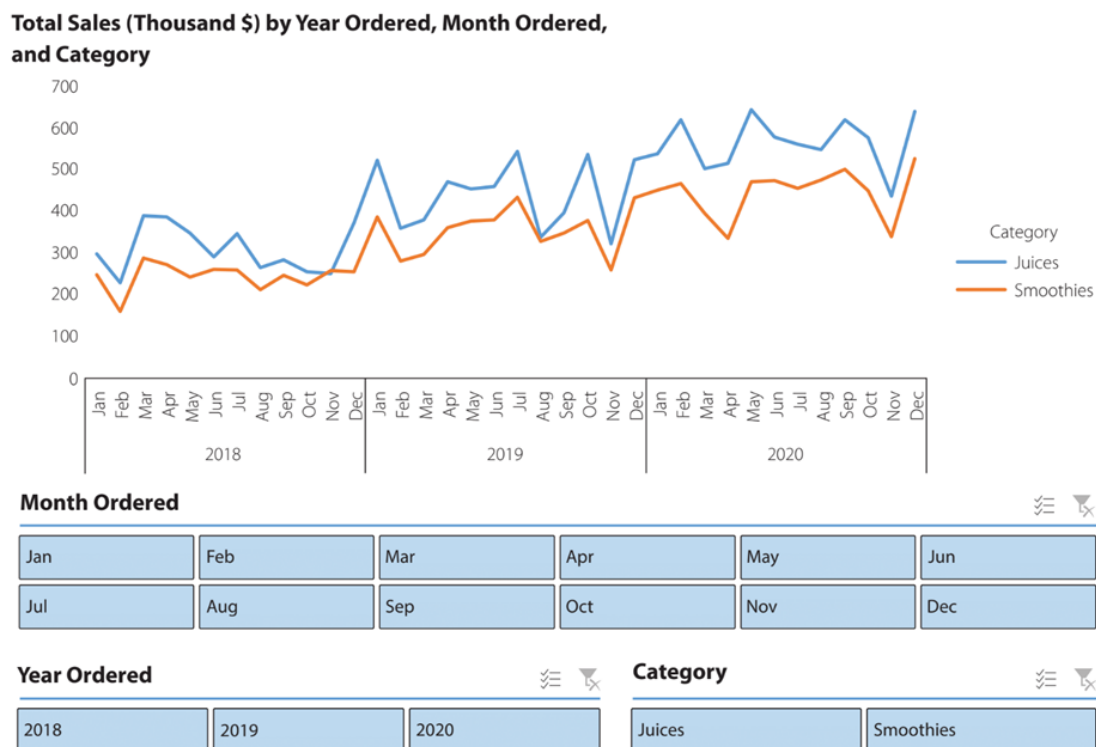
Open the *EJBChart1* data file and complete the following steps to build the line chart:

- Step 1.
 - In the file *EJBChart1*, select any cell in the *EJBData* table in the Data worksheet
 - Click the Insert tab on the Ribbon

- In the Charts group click PivotChart and then select PivotChart & PivotTable
- Step 2.
 - When the Create PivotTable dialog box appears:
 - Under Choose the data that you want to analyze, choose Select a table or range and enter EJBData in the Table/Range: box
 - Under Choose where you want the PivotTable report to be placed, select New Worksheet
 - Click OK
- Step 3.
 - Change the name of the new worksheet to Chart2
 - We next create the line chart of total sales by month and year by category in the Chart2 worksheet.
- Step 4.
 - In the PivotChart Fields task pane:
 - Drag the \$ Sales field to the Values area
 - Drag the Year Ordered and Month Ordered fields to the Axis (Categories) area
 - Drag the Category field to the Legend (Series) area
 - Use the Value Field Settings from the drop-down menu of \$ Sales to select Sum for the \$ Sales field
- Step 5.
 - Click the chart, click the Design tab on the PivotChart Tools Ribbon, click the Change Chart Type button in the Type group, click Line on the left pane, and select Line in the right pane
 - Click OK
- Step 6.
 - Right click the vertical axis of the chart and click Format Axis to open the Format Axis task pane
 - Click Axis Options and then click the Axis Options button
 - Under Axis Options, enter 0 in the Minimum box
 - In the Display Units box, select Thousands
 - Click on the vertical axis title of “Thousands” that has automatically been generated on the chart and press the Delete key
- Step 7.
 - Right click the horizontal axis and click Format Axis to open the Format Axis task pane
 - Click Axis Options and then click the Axis Options button
 - Click Labels, select Specify interval unit, and enter 1
- Step 8.
 - Right click any Field Button (such as month ordered) and select Hide All Field Buttons on Chart
- Step 9.
 - Click the PivotChart, click the Insert tab on the Ribbon, and click Slicer in the Filters group
- Step 10.
 - When the Insert Slicers dialog box appears:

- Select the check boxes for Category, Year Ordered, and Month Ordered
 - Click OK
- Step 11.
 - Click each of the slicers and use the tools in the Slicer tab on the Ribbon to format the slicer as appropriate
 - Drag the slicers to the positions they should occupy and the dashboard and resize the slicers accordingly

Edit the chart to improve its appearance. The result should look similar to the following figure.



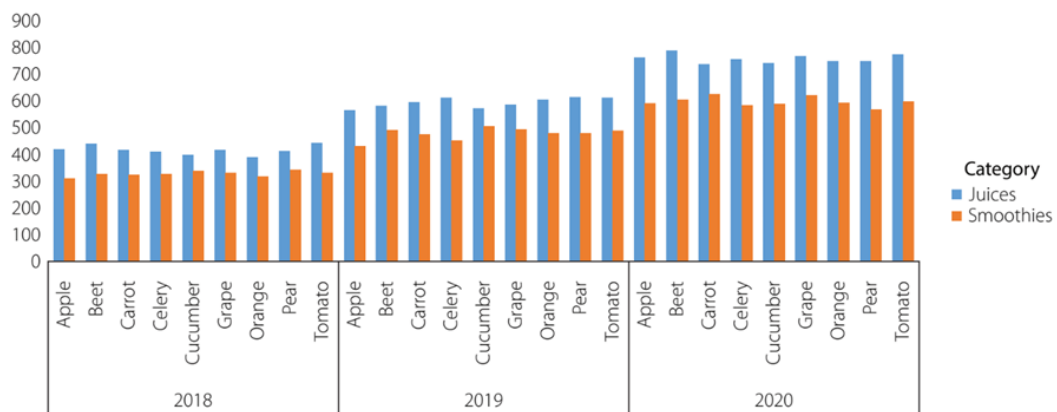
To create the clustered column chart of total sales across year ordered and flavour by category, complete the following steps:

- Step 1.
 - In the file EJBChart1, select any cell in the EJBData table in the Data worksheet
 - Click the Insert tab on the Ribbon
 - In the Charts group click PivotChart and then select PivotChart & PivotTable
- Step 2.
 - When the Create PivotTable dialog box appears:
 - Under Choose the data that you want to analyse, select Select a table or range and enter EJBData in the Table/Range: box
 - Under Choose where you want the PivotTable report to be placed, select New Worksheet

- Click OK
- Step 3.
 - Change the name of the new worksheet to Chart3
- Step 4.
 - In the PivotChart Fields task pane:
 - Drag the \$ Sales field to the Values area
 - Drag the Year Ordered and Flavor fields to the Axis (Categories) area
 - Drag the Category field to the Legend (Series) area
 - Use the Value Field Settings from the drop-down menu of \$ Sales to select Sum for the \$ Sales field
- Step 5.
 - Click the chart, click the Design tab on the PivotChart Tools Ribbon, click the Change Chart Type button in the Type group, click Column on the left pane, and select Clustered Chart in the right pane
 - Click OK
- Step 6.
 - Right click the vertical axis of the chart and click Format Axis to open the Format Axis task pane
 - Click Axis Options and then click the Axis Options button
 - Under Axis Options, enter 0 in the Minimum box
 - In the Display Units box, select Thousands
 - Click on the vertical axis title of “Thousands” that has automatically been generated on the chart and press the Delete key
- Step 7.
 - Right click any Field Button (such as month ordered) and select Hide All Field Buttons on Chart
- Step 8.
 - Click the PivotChart, click the Insert tab on the Ribbon, and click Slicer in the Filters group
- Step 9.
 - When the Insert Slicers dialog box appears:
 - Select the check boxes for Category, Year Ordered, and Flavor
 - Click OK
- Step 10.
 - Click each of the slicers and use the tools in the Slicer tab on the Ribbon to format the slicer as appropriate
 - Drag the slicers to the positions they should occupy and the dashboard and resize the slicers accordingly

Edit the chart to improve its appearance. The result should look similar to the following figure.

Total Sales (Thousand \$) by Year Ordered, Flavor, and Category



Flavor

Apple	Beet	Carrot	Celery	Cucumber
Grape	Orange	Pear	Tomato	

Year Ordered

2018	2019	2020
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Category

Juices	Smoothies
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To create a clustered bar chart of total sales across distribution centre by year ordered, complete the following steps:

- Step 1.
 - In the file EJBChart1, select any cell in the EJBData table in the Data worksheet
 - Click the Insert tab on the Ribbon
 - In the Charts group click PivotChart and then select PivotChart & PivotTable
- Step 2.
 - When the Create PivotTable dialog box appears:
 - Under Choose the data that you want to analyse, choose Select a table or range and enter EJBData in the Table/Range: box
 - Under Choose where you want the PivotTable report to be placed, select New Worksheet
 - Click OK
- Step 3.
 - Change the name of the new worksheet to Chart4

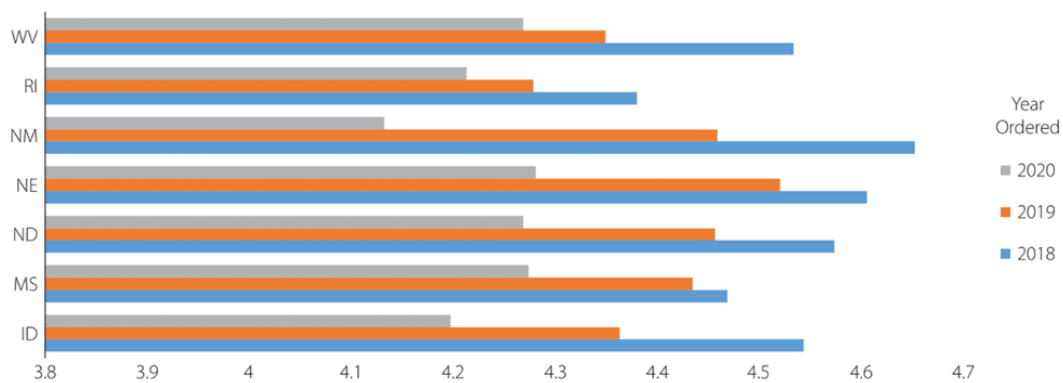
Next, the clustered bar chart of total sales across distribution centre by year ordered will be created in the Chart4 worksheet.

- Step 4.
 - In the PivotChart Fields task pane:

- Drag the Time to Deliver field to the Values area
 - Drag the Distribution Center field to the Axis (Categories) area
 - Drag the Year Ordered field to the Legend (Series) area
 - Use the Value Field Settings from the drop-down menu of Time to Deliver to select Average for the Time to Deliver field
- Step 5.
 - Click the chart, click the Design tab on the PivotChart Tools Ribbon, click the Change Chart Type button in the Type group, click Bar on the left pane, and select Clustered Bar in the right pane
 - Click OK
- Step 6.
 - Right click the vertical axis and click Format Axis to open the Format Axis pane
 - Click Axis Options and then click the Axis Options button
 - Click Labels, click Specify interval unit, and enter 1 into the Specify interval unit box
 - Click OK
- Step 7.
 - Right click any Field Button (such as year ordered) and select Hide All Field Buttons on Chart
- Step 8.
 - Click the PivotChart, click the Insert tab on the Ribbon, and click Slicer in the Filters group
- Step 9.
 - When the Insert Slicers dialog box appears:
 - Select the check boxes for Distribution Center and Year Ordered
 - Click OK
- Step 10.
 - Click each of the slicers and use the tools in the Slicer tab on the Ribbon to format the slicer as appropriate
 - Drag the slicers to the positions they should occupy and the dashboard and resize the slicers accordingly

Edit the chart to improve its appearance. The result should look similar to the following figure.

**Average Time to Delivery by Distribution Center and Year
Ordered**



Distribution Center

ID	MS	ND	NE	NM	RI	WV
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Year Ordered

2018	2019	2020
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To create the PivotTable of total dollar sales and average time to deliver across category and flavour by distribution centre and year, month, and day ordered, complete the following steps:

- Step 1.
 - In the file *EJBCharts*, create a new worksheet and name this worksheet Dashboard

Step 2 changes the fill colour of cells A1:Z100 to a dark colour to provide a contrasting background for the dashboard.

- Step 2.
 - Select cells A1:Z100 Click the Home tab on the Ribbon and in the Font group, click the Fill Color button and select a dark blue colour
- Step 3.
 - In the file *EJBCharts*, select any cell in the *EJBData* table in the Data worksheet
 - Click the Insert tab on the Ribbon and select PivotTable in the Tables group
- Step 4.
 - When the Create PivotTable dialog box appears:
 - Under Choose the data that you want to analyse, click Select a Table or Range and enter *EJBData* in the Table/Range: box
 - Under Choose where you want the PivotTable report to be placed, select Existing Worksheet and enter Dashboard!B4
 - Click OK

- Step 5.
 - In the Dashboard worksheet
 - Click the empty PivotTable to open the PivotTable Fields task pane
- Step 6.
 - In the PivotTable Fields pane:
 - Drag the \$ Sales and Time to Deliver fields to the Values area
 - Drag the Year Ordered, Month Ordered, Day Ordered, and Distribution Center fields to the Filters area
 - Drag the Category field to the Columns area (make sure this is the first field listed in the Columns area)
 - Drag the Flavor field to the Rows area
 - Click the drop down arrow next to Time to Deliver in the Values area, click Value Field Settings and change the Summarize value field by to Average
 - Click OK
- Step 7.
 - Right click on any cell in the PivotTable and select PivotTable Options
- Step 8.
 - When the PivotTable Options dialog box opens:
 - Click the Layout & Format tab
 - Deselect the check box for Autofit column widths on update

The result should look similar to the following figure.

	A	B	C	D	E	F	G
1							
2							
3							
4		Year Ordered	(All)				
5		Month Ordered	(All)				
6		Day Ordered	(All)				
7		Distribution Center	(All)				
8							
9							
10		Column Labels					
11		Row Labels	Sum of \$ Sales	Average of Time to Delivery	Smoothies Sum of \$ Sales	Average of Time to Delivery	Total Sum of \$ Sales
12		Apple	\$1,746,990	4.4642	\$1,330,694	4.3665	\$3,077,684
13		Beet	\$1,805,180	4.3649	\$1,418,969	4.4392	\$3,224,149
14		Carrot	\$1,748,337	4.3631	\$1,423,311	4.4293	\$3,171,647
15		Celery	\$1,774,726	4.4234	\$1,360,251	4.2735	\$3,134,977
16		Cucumber	\$1,710,376	4.3108	\$1,429,580	4.3693	\$3,139,956
17		Grape	\$1,770,301	4.4432	\$1,441,864	4.3115	\$3,212,164
18		Orange	\$1,740,004	4.4367	\$1,387,955	4.3809	\$3,127,959
19		Pear	\$1,771,377	4.4047	\$1,389,352	4.4240	\$3,160,728
20		Tomato	\$1,823,193	4.3985	\$1,414,450	4.3529	\$3,237,643
21		Grand Total	\$15,890,484	4.4012	\$12,596,425	4.3719	\$28,486,909
22							

The next step is to assemble the data dashboard. This requires that you consider the relative positioning of each of the charts on the dashboard. The objective is to minimize eye travel by putting charts that are likely to be used together in proximity to each other. For example:

- Chart 1 and Chart 3 display sales history by year
- Chart 1 and Chart 4 display analyses at the distribution centre level
- Chart 2 and Chart 3 display sales at category level

Easy access is required for the table of total sales across category and flavour by distribution centre and date ordered. A suggested configuration is shown below.



To complete construction of the EJB data dashboard in this layout, complete the following steps:

- Step 1.
 - Click the PivotChart in Chart1 worksheet
- Step 2.
 - Click the Analyze tab on the PivotChart Tools Ribbon, and click the Move Chart button in the Actions group
- Step 3.
 - When the Move Chart dialog box appears:
 - Select Object in: and choose Dashboard
 - Click OK
- Step 4.
 - In the Chart1 worksheet, click on the Distribution Center slicer
 - Click the Home tab on the Ribbon
 - Click Cut in the Clipboard group
- Step 5.
 - In the Dashboard worksheet
 - Click the Home tab on the Ribbon
 - Click Paste
- Step 6.
 - Position the PivotChart and slicer where you want them on the dashboard
- Step 7.

- Repeat Steps 1–6 to move the charts and slicers in Chart2, Chart3, and Chart4 worksheets to the Dashboard worksheet and position these objects where you want them on the dashboard
- Step 8.
 - Double click one of the PivotCharts in the Dashboard worksheet to open the Format Chart Area task pane
 - Click Chart Options, then click the Size & Properties button
 - Click Properties, then select Don't move or size with cells
- Step 9.
 - Repeat Step 8 for the remaining charts on the Dashboard worksheet
- Step 10.
 - Right click one of the slicers in the Dashboard worksheet, then select Size and Properties to open the Format Slicer dialog box
 - Click Properties, then select Don't move or size with cells
- Step 11.
 - Repeat Step 10 for the remaining slicers on the Dashboard worksheet
- Step 12.
 - Right click any chart on the Dashboard worksheet and select PivotChart Options
- Step 13.
 - When the PivotTable Options dialog box opens:
 - Click the Layout & Format tab
 - Deselect the check box for Autofit column widths on update
 - Click OK
- Step 14.
 - Adjust the positioning and size of the charts and slicers so they align as desired
- Step 15.
 - Select the View tab on the Ribbon, and in the Show group deselect the check box for Gridlines

At this point if you use a slicer to filter out some series on a chart, the colours assigned to the remaining series may change. Another issue is that, the dashboard uses blue and orange to differentiate between new and existing customers, juices and smoothies, and 2018 and 2019. It would be better to use different sets of colours to consistently differentiate between each of the series across charts.

To address these issues, complete the following steps:

- Step 16.
 - Click the File tab on the Ribbon, then select Options
- Step 17.
 - When the Excel Options dialog box appears:
 - Click Advanced

- Under Chart, select the check boxes for Properties follow chart data point for all new workbooks and Properties follow chart data point for current workbook
- Step 18.
 - Assign the specific colours below to different chart elements by right clicking on that chart element, clicking Fill and selecting More Fill Colors
 - Blue to existing customers
 - Orange to new customers
 - Purple to juices and Red to smoothies
 - Brown to 2018, Pink to 2019, and Green to 2020
- Step 19.
 - Change the fill colour of cells A1:AL80 to dark blue to create a contrasting background
- Step 20.
 - Type Espléndido Jugo y Batido, Inc. Sales Dashboard into cell A2
 - Change the font to white 48 pt. Brush Script MT font
- Step 21.
 - Select cells A2:AF2, select the Home tab on the Ribbon, and in the Alignment group click Merge & Center

With some further editing of the table, the result should look similar to the following figure.



In addition, if desired you can change the style and settings for any slicers by clicking the slicer, clicking Options on the Ribbon, then using the various tools such as those in the Slicer Styles group.

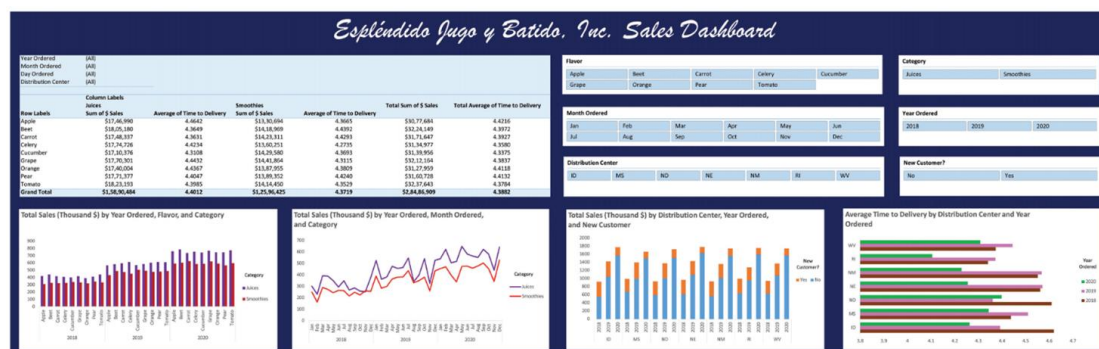
If you want to create a data dashboard with a single slicer that controls multiple related charts, complete the following steps:

- Step 1.
 - Delete all but one of the Distribution Center slicers from the EJB dashboard
- Step 2.
 - Select the remaining Distribution Center slicer on the Dashboard worksheet
 - Click the Slicer tab on the Ribbon
 - Click the Report Connections button in the Slicer group
- Step 3.
 - Select both the PivotTables created in the Chart1 worksheet and the Chart4 worksheet
 - Click OK

Because the PivotCharts created in the Chart1 and Chart4 worksheets are linked to the corresponding PivotTables in the Chart1 and Chart4 worksheets, the single Distribution Center slicer now controls the value(s) of Distribution Center that are displayed in both of these charts.

- Step 4.
 - Repeat Steps 2 and 3 to link the PivotTables created in the Chart2 and Chart3 worksheets to a single Category slicer and to link the PivotTables created in the Chart1, Chart2, Chart3, and Chart4 worksheets to a single Year Ordered slicer

The elements of the dashboard can now be rearranged to make the dashboard more visually appealing and easier to read whilst still meeting the original criteria for the placement of the charts. This can be seen in the following figure.



To protect the dashboard from being altered by users, complete the following steps:

- Step 1.
 - Right click any slicer on the Dashboard worksheet, then click Size and Properties
- Step 2.
 - When the Format Slicer task pane appears:
 - Under Position and Layout select the check box for Disable resizing and moving
 - Under Properties select Don't move or size with cells and deselect the check box for Locked
- Step 3.
 - Repeat Steps 1 and 2 for each remaining slicer on the Dashboard worksheet
- Step 4.
 - Select the range of cells that contains the PivotTable in the Dashboard worksheet (this is cells B4:H21 on our Dashboard worksheet)
 - Click the Review tab on the Ribbon
 - In the Protect group, click the Allow Edit Ranges button
- Step 5.
 - When the Allow Users to Edit Ranges dialog box appears:
 - Click the New button
- Step 6.
 - When the New Range dialog box appears:
 - Enter DashboardPivotTable in the Title: box
 - Click OK
- Step 7.
 - When the Allow Users to Edit Ranges dialog box returns:
 - Click the Protect Sheet button
- Step 8.
 - When the Protect Sheet dialog box opens:
 - Deselect the check box for Select locked cells
 - Select the check box for Use PivotTable & PivotChart
 - Enter a password in the Password to unprotect sheet: box
 - Re-enter the same password in the Re-enter password to proceed. box of the Confirm Password dialog box dialog box (recall that we used the password TRIAL to protect the dashboard in the file EJBDashboard2)
 - Click OK
- Step 9.
 - Select the Chart1 worksheet tab, then hold the Ctrl key down and select the Chart2, Chart3, Chart4, and Data worksheet tabs so all worksheet tabs in the EJB2Dashboard file except the Dashboard worksheet are selected
 - Right click the Chart1 worksheet tab and select Hide